



BANGLADESH UNIVERSITY OF PROFESSIONALS (BUP)

Project Proposal

Tentative Title: Light Following Robot

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I. INTRODUCTION

Automation has become prevalent in various fields, offering advantages such as reduced manual operations, enhanced flexibility, reliability, and increased accuracy. Modern systems, especially in electronics, benefit significantly from automated control systems. Unmanned systems, like drones, play a crucial role in minimizing risks to human life, particularly in warfare. Recent advancements in robotics have spurred a demand for self-sustaining automated systems capable of efficient task detection and maneuvering.

In both global contexts and Bangladesh, challenges persist across various sectors, encompassing manufacturing, agriculture, construction, and industry. These sectors are fundamental to economic growth and societal development but grapple with systemic issues that hinder their efficiency and effectiveness. In various industrial sectors, such as manufacturing, textiles, and agro-processing, productivity issues and labor shortages hinder growth and competitiveness. Manual labor reliance, coupled with outdated production techniques, restricts the industries' potential for innovation and modernization. Moreover, in the global marketplace, where competition is fierce, industries struggle to maintain their edge due to inefficiencies in production processes and limited access to advanced technologies. In the manufacturing sector, labor shortage and the lack of automation infrastructure lead to notable inefficiencies. Traditional manufacturing processes are often slow, error-prone, and costly, resulting in decreased competitiveness and profitability for businesses. Similarly, the agricultural sector faces significant challenges, exacerbated by labor shortages and outdated farming practices. In Bangladesh, where agriculture is a vital component of the economy, the sector struggles with insufficient mechanization, limited access to modern farming equipment, and a shortage of skilled labor. As a result, farmers experience difficulties in optimizing their yields, mitigating crop losses, and meeting the growing food demand of the population. The construction industry confronts its

own set of obstacles, including labor shortages, safety concerns, and project delays. Manual construction methods prevail, contributing to inefficiencies, slower project completion times, and increased risks to worker safety. Bangladesh is facing tremendous accident issues at construction sites, and it gets larger size every year. The statistics of accidents at construction sites show us a picture that the Bangladeshi construction sector is in most critical situation that needs a huge and fast overcome from the current frequent accidental level [1].

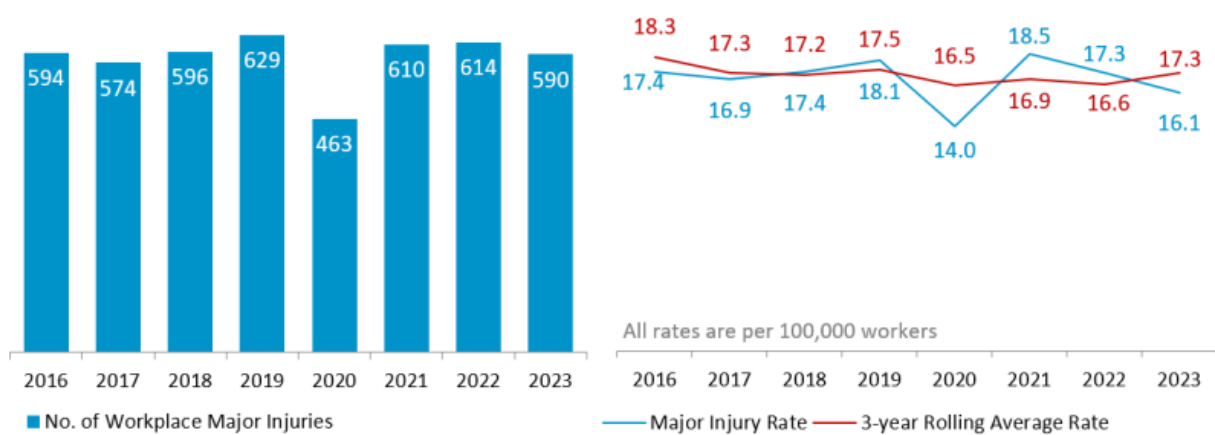


Figure I: Number and rate of workplace major injuries, 2016-2023 [2]

A light-following robot has the potential to address several challenges and contribute to the development of both Bangladesh and the world. Firstly, the implementation of a light-following robot could help in addressing the problem of labor shortages in various industries, such as manufacturing, agriculture, and construction. The robot could perform tasks that are repetitive, dangerous, or require a high level of precision, thus freeing up human workers to focus on more complex tasks. This could lead to increased productivity and efficiency in these industries, which could have a positive impact on the economy of Bangladesh and other countries. Secondly, the development and implementation of a light-following robot could help in reducing the risk of accidents and injuries in the workplace. The robot could be designed to work in

hazardous environments, such as mines, construction sites, or chemical plants, where human workers are at risk of exposure to harmful substances or accidents. By reducing the risk of accidents and injuries, the robot could help in improving the working conditions for human workers and reducing the costs associated with workplace accidents and injuries. Thirdly, the light-following robot could have a positive impact on the environment. By reducing the need for human workers to travel to and from work, the robot could help in reducing the carbon footprint of industries. Additionally, the robot could be designed to perform environmentally friendly tasks, such as recycling or waste management [3][4]. Existing light-following robot designs globally often struggle with responsiveness to varying light conditions and fail to operate effectively in diverse environments. This limitation impedes their widespread adoption and diminishes their potential impact on automation and robotics [5].

In summary, this project has the potential -

- To address labor shortages
- To improve working conditions
- To reduce the carbon footprint of industries and perform eco-friendly tasks

II. LITERATURE REVIEW

Light detecting or following robots is intertwined with the advancement of robotics and sensor technology. The development of light-detecting and following robots boasts a rich history spanning several decades. This evolution has been intricately linked to advancements in both sensor technology and control algorithms. While the concept of robots following light traces back to the mid-20th century, significant developments have occurred in recent decades.

Grey Walter's light-sensitive robot, also known as *Machina Speculatrix*, was a pioneering creation in the field of robotics. Developed by British neuroscientist and roboticist William Grey Walter in the late 1940s and early 1950s, these robots were among the earliest examples of autonomous mobile robots.

Grey Walter's robots had a tortoise-like appearance with a simple body structure made of metal and plastic. The robots had light sensors, batteries, and electronic circuitry for basic behavioral control. The primary feature of these robots was their ability to respond to light stimuli. They had light sensors mounted on their "heads" that allowed them to detect light sources in their environment. The robots demonstrated autonomous movement by exhibiting behaviors such as seeking out light sources and moving towards or away from them based on their intensity. While their behaviors seemed complex for the time, the underlying control mechanisms were relatively simple compared to modern robotics [6].

Advancements in sensor technology during the 1980s-1990s contributed to the development of light-detecting and light-following robots. Improvements in photodetectors, photodiodes, and phototransistors enabled robots to detect and respond to light sources more accurately and reliably. Rodney Brooks, a roboticist and artificial intelligence researcher, made significant contributions to behavior-based robotics during this period [7]. His work focused on simple, reactive robots that could exhibit complex behaviors through interactions with their environment. Brooks' research laid the foundation for the development of light-following robots with robust and adaptive control systems. Although Brooks' research did not focus exclusively on light-following robots, his work laid the groundwork for the development of such robots by promoting a paradigm shift in robotics toward more robust and adaptive control systems. His ideas continue to influence the design and implementation of autonomous systems in various domains, including robotics, artificial intelligence, and autonomous vehicles [8].

Steve Furber and Sophie Wilson's primary contributions are in the fields of computer architecture and microprocessor design, particularly with the development of the BBC Microcomputer System and the ARM architecture. Their work has focused on digital computing systems rather than light detecting technology. Although ARM architecture is widely used in embedded systems, IoT devices, and mobile platforms, where sensors, including light sensors, are commonly integrated for various functionalities [9].

In the modern era, multiple attempts were made in this sector. Light-following robots equipped with PID (Proportional-Integral-Derivative) controllers use feedback control loops to adjust their movement based on light sensor readings. While PID controllers can improve the stability and precision of light-following robots, tuning the controller parameters can be challenging. These robots may exhibit oscillations or overshoot in response to sudden changes in light intensity.

Some commercially available light-following robot toys marketed for entertainment or educational purposes may have faced criticisms for their limited functionality, reliability issues, or lack of robustness. These robots often target a broad consumer audience and may prioritize affordability and simplicity over advanced features or performance.

Despite advancements in sensor technology, microcontrollers, and AI algorithms, modern light-following robots may still struggle with certain environmental conditions, such as highly variable lighting, complex backgrounds, or reflective surfaces. Technical constraints may limit the accuracy, speed, or versatility of light detection and navigation capabilities.

III. METHODOLOGY

In this project of a Light-following robot, we are going to use the Easier Pro Development Board, four light-dependent sensors, two TT gear motors, a car chassis, two wheels and a caster wheel as our main components. There are also some sub-components to complete the project.

Firstly, we are going to attach the development board to the car chassis. Then we connect four sensors and a 5V DC power supply to the development board. There is an intensity regulator attached to the sensor board. We fix the intensity to take input correctly through the sensors. We connect the motors with the “sig-1” pin, so that when the sensors sense the light rays the “sig-1” pin output is 1 and the motors start.

Secondly, two sensors are placed on the left and right, and the other two motors are placed in front of the car chassis. The sensor placed on the left is connected to the motor placed on the right and the sensor placed on the right is connected to the motor placed on the left. The sensors in front are connected to both motors.

When the left sensor senses the light rays, the right motor starts rotating, and when the right sensor, the left motor starts rotating. When the front sensors sense the light rays both the motor starts rotating.

Components Description:

1. Easier Pro Development Board: It is the main component of the project. It combines all the input and output signals. Some input pins take inputs from different sensors. The output pins can be connected to different output devices. There is a processing device known as a “Relay”. It is a switching device. There is a “ULV-2003” IC that works as a Relay driver.

2. Light Sensor: This light sensor is an LDR (Light Dependent Resistor). These sensors change their resistance based on the amount of light falling on them. More light decreases the resistance, while less light increases it. Light sensors are employed in robotics for tasks such as line following, obstacle detection, or light-based communication. Light sensors typically utilize semiconductor materials that generate electron-hole pairs when exposed to light. The number of electron-hole pairs produced is directly proportional to the intensity of incident light. This change in electrical properties (current, voltage, resistance) is then measured and processed to determine the light level.

3. TT Gear Motor: The TT gear motor contains a small electric motor that converts electrical energy (from a battery or power supply) into mechanical energy. When electricity flows through the motor, it causes the motor's shaft to rotate. The gearbox consists of gears (typically metal) that control the speed and torque (rotational force) of the motor's output. Gears in the gearbox mesh together to transmit power from the motor to the output shaft.

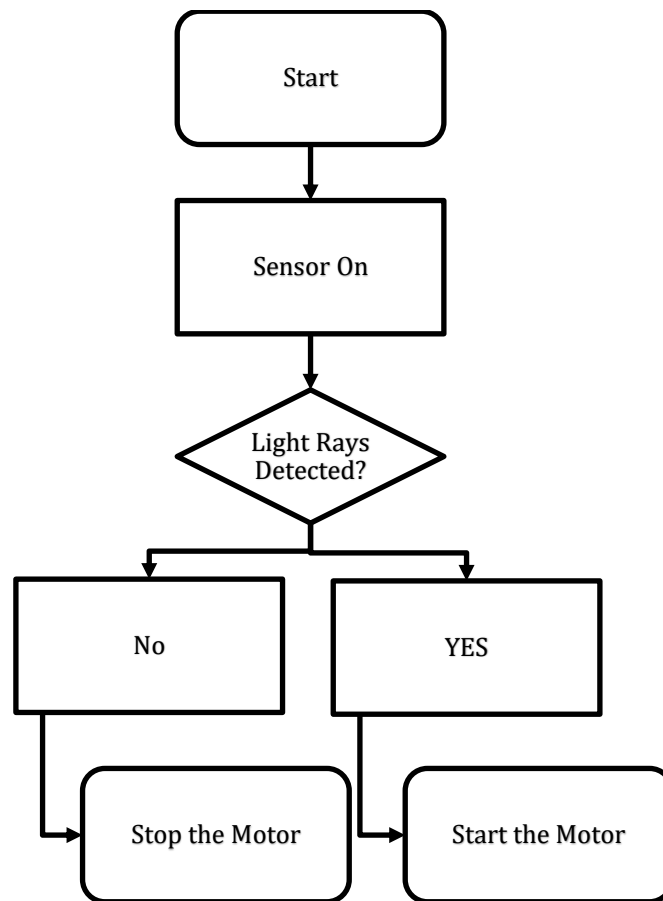


Figure II: Process Flowchart

When the light sensor (LDR) senses the light rays, it sends a signal to the output pin through the processor. The processor driver transforms the signal into an electric high voltage. And the processor works like a switch so the motor starts rotating.

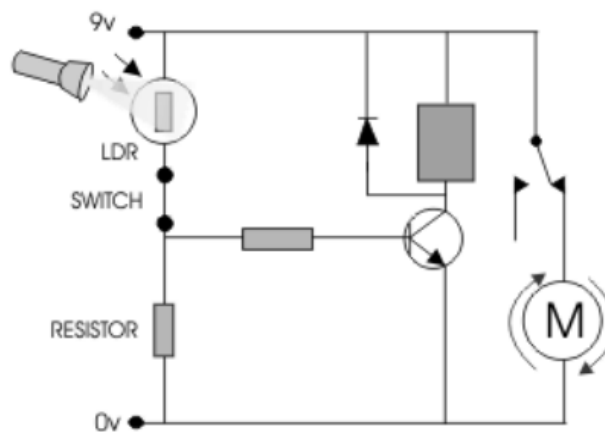


Figure III: Working Method of Proposed Model [10]

IV. EXPECTED RESULTS

The implementation of light-following robots promises immediate benefits in efficiency, safety, and productivity across sectors like manufacturing, agriculture, and construction. In the long term, sustained deployment can drive technological innovation, enhance productivity, and reshape employment patterns towards higher-skilled roles. This fosters economic development and positions Bangladesh as a leader in adopting advanced technologies. Moreover, advancements in this technology can lead to breakthroughs in healthcare, security, transportation, and logistics, contributing to global progress. In summary, adopting light-following robots offers significant potential for positive change, locally and globally, with implications for economic growth, innovation, and societal well-being.

V. DESCRIPTION OF THE WORK PLAN

This project is divided into multiple interrelated stages that follow a top-down approach. The initial four weeks will be dedicated to researching existing light-following robot projects. The collected data will then be compiled and analyzed within a two-week timeframe. Component acquisition for the robot's construction will take approximately one week. Building the model itself is expected to be the most time-consuming phase, allocated four weeks for assembly and integration. Following successful development, a week will be dedicated to reporting the project's results, with an additional week for final report preparation.

Table I
WEEKLY WORK PLAN FOR THE PROJECT

Work Plan	Weeks											
Research on Previous Projects												
Compilation of the Proposal												
Equipment Collection												
Model Development & Reporting												
Final Project Submission												

VI. BUDGET ESTIMATION & REPORT

The components used to develop the core hardware of the system are listed in Table II & the budget report in this regard is shown in Figure I.

TABLE II
LIST OF COMPONENTS AND THE COST OF DEVELOPMENT

SL.	Components	Quantity	Unit Price (TK)	Total Price (TK)
1	Light Sensors	4	80	320
2	Easier Pro-development Board	1	520	520
3	3-Pin Jumper Wires	4	30	120
4	Robot PVC Chassis with Wheels and Motors	1	430	430
	Subtotal			1390
5	Contingency		10%	139
Total				1529

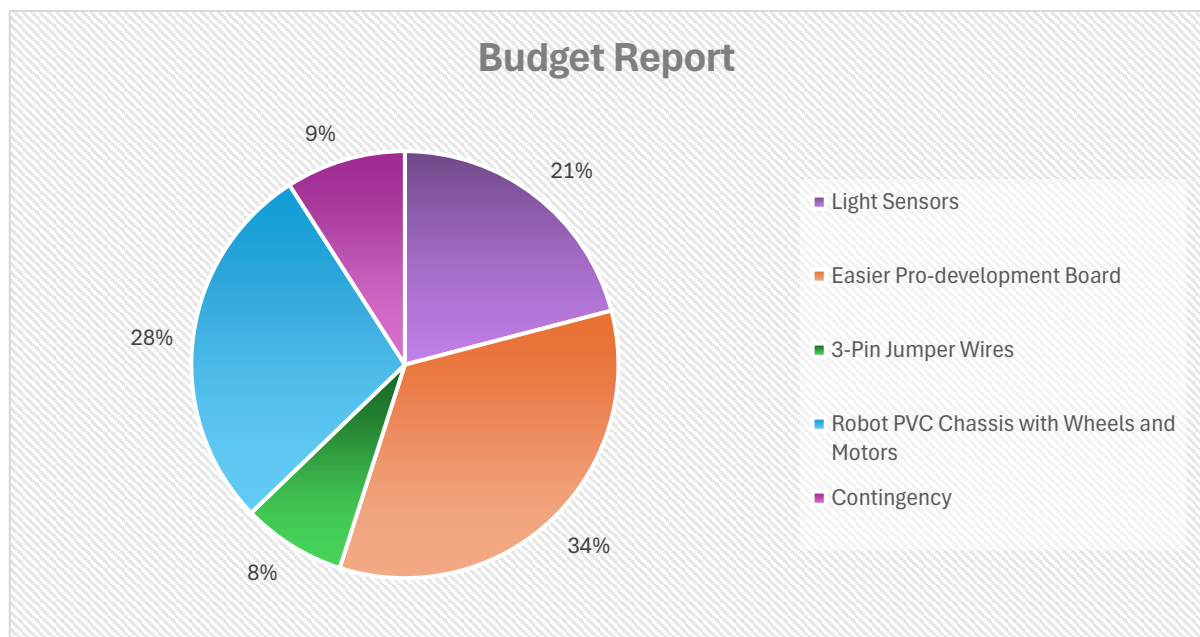


FIGURE IV: Pie-chart showing the Budget Report

The estimated budget for the Light Following Robot project is approximately 1529 Taka (TK). The bulk of the expenses are allocated to materials and components, with the robot chassis with wheels and motors being the single largest cost factor. To account for unforeseen situations, a contingency fund of 10% (approximately 139 TK) has been included. This could cover additional hardware (breadboard, batteries, fasteners), replacement parts, shipping costs, or software development tools (if programming is required). By incorporating this buffer, the project can adapt to unexpected needs and stay within the proposed budget.

References:

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